

CSE-415: Human Computer Interaction

COURSE INFORMATION						
Course Code		: CSE-415	Lecture Contact Hours		: 3.00	
Course Title		: Human Computer Interaction	Credit Hours		: 3.00	
PRE-REQUISITE						
Course Code: Nil Course Title: Nil						
CURRICULUM STRUCTURE						
Outcome Based Education (OBE)						
RATIONALE						
The Human Computer Interaction course covers the foundations of Human Computer Interaction (HCI), a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them.						
OBJECTIVE						
1. To understand the definitions and foundations of the HCI domain. 2. To design interfaces and interactive solutions using user-centered techniques. 3. To apply evaluation methods, quality factors, and data analysis techniques. 4. To explore research frontiers of HCI, including universal design, responsive design and pervasive computing.						
LEARNING OUTCOMES & GENERIC SKILLS						
No.	Course Learning Outcome (Upon completion of the course, the students will be able to)	Bloom's Taxonomy	CP	CA	KP	Assessment Methods
CO1	Understand and applying the fundamentals of HCI and Interaction design.	C1- C3	1		3	T, F
CO2	Analyse the focused users and system requirements, and to design different kind of UIs and Interaction systems for building intuitive usable software solutions.	C4, C6	2		4, 5	T, MT
CO3	Apply (design) evaluation methods for assuring the enhanced usability including effectiveness, efficiency and satisfaction	C4	1		8	F
CO4	Develop the communication skill by presenting topics on HCI.	A2		1		Pr
(CP- Complex Problems, CA-Complex Activities, KP-Knowledge Profile, T – Test; PR – Project; Q – Quiz; ASG – Assignment; Pr – Presentation; R - Report; F – Final Exam, MT- Mid Term Exam)						
COURSE CONTENT						
Introduction to HCI and Interaction design: HCI, Interaction design, The process of design, User focus, Scenarios, Navigation design, Screen design and layout, Iteration and prototyping. HCI in the software process: The software life cycle, Usability engineering, Iterative design and prototyping, Design rationale. Design rules: Principles to support usability, Standards, Guidelines, Golden rules and heuristics, HCI patterns. Evaluation techniques: What is evaluation? What, why, and when to evaluate, Goals of evaluation, Evaluation through expert analysis, Evaluation through user participation, Choosing an evaluation method. Evaluation paradigms and techniques, The D E C I D E framework to guide evaluation. Observing users: Participant observation, ethnography, Data collection, and Analyzing, interpreting and presenting data, Qualitative analysis, Feeding the findings back into design. Asking users and experts: Interviews, Questionnaires, Inspections, walkthroughs. Universal design: Universal design principles, multi-modal interaction, designing for diversity. Task analysis: Differences between task analysis and other techniques, Task decomposition, Knowledge-based techniques, Entity-relationship-based techniques, Sources of information and data collection, Uses of task analysis. Modeling rich interaction: Status-event analysis, Rich contexts, Low intention and sensor-based interaction. Ubiquitous computing and augmented realities: Ubiquitous computing applications research, virtual and augmented reality, Information and data visualization. Hypertext, multimedia and the world wide web: Understanding						

hypertext, Finding things, Web technology and issues, Static web content, Dynamic web content

SKILL MAPPING

No.	Course Learning Outcome	PROGRAM OUTCOMES (PO)											
		1	2	3	4	5	6	7	8	9	10	11	12
CO1	Understand and applying the fundamentals of HCI and Interaction design.	H											
CO2	Analyse the focused users and system requirements, and to design different kind of UIs and Interaction systems for building intuitive usable software solutions.		M	H									
CO3	Apply (design) evaluation methods for assuring the enhanced usability including effectiveness, efficiency and satisfaction				H								
CO4	Develop the communication skill by presenting topics on HCI.										L		

(H – High, M- Medium, L-low)

JUSTIFICATION FOR CO-PO MAPPING

Mapping	Level	Justifications
CO1-PO1	High	Acquire a good level of knowledge regarding HCI, especially on interaction design and evaluation through the fundamental concept of HCI like interaction design, design process, design principles, universal design, rich interaction, etc.
CO2-PO2, PO3	Medium, High	Understand how to profile the focused users and reveal the user requirements from usability perspective; as well as the interaction design techniques, rules, and principles for design the software systems including the ubiquitous computing, augmented realities and rich interfaces, by acquiring the knowledge regarding the ability to understand user-profile and design any usable information systems.
CO3-PO3	High	Conduct an in-depth usability/ design evaluation to (re)design any information system in order to assure the enhanced usability and user-experience by having a good level of familiarity with different evaluation methodologies.
CO5-PO10	Low	Develop communication skills through participating in presentation.

TEACHING LEARNING STRATEGY

Teaching and Learning Activities	Engagement (hours)
Face-to-Face Learning	
Lecture	42
Practical / Tutorial / Studio	-
Student-Centred Learning	-
Self-Directed Learning	
Non-face-to-face learning	42
Revision	21
Assessment Preparations	21
Formal Assessment	
Continuous Assessment	2
Final Examination	3
Total	131

TEACHING METHODOLOGY

Lecture and Discussion, Co-operative and Collaborative Method, Problem Based Method

COURSE SCHEDULE			
Week	Lecture	Topics	Assessment Methods
1	1	Introduction to HCI and Interaction design: HCI, Interaction design, The process of design	Class Test 1
	2	User focus, Scenarios, Navigation design	
	3	Screen design and layout, Iteration and prototyping.	
2	4	HCI in the software process: The software life cycle,	
	5	Usability engineering	
	6	Iterative design and prototyping, Design rationale.	
3	7	Design rules: Principles to support usability, Standards	
	8	Guidelines, Golden rules and heuristics	
	9	HCI patterns	
4	10	Evaluation techniques: What is evaluation? What, why, and when to evaluate, Goals of evaluation	Class Test 2
	11	Evaluation through expert analysis	
	12	Evaluation through user participation, Choosing an evaluation method.	
5	13	Evaluation paradigms and techniques,	
	14	The D E C I D E framework to guide evaluation.	
	15	The D E C I D E framework to guide evaluation. (Contd.)	
6	16	Observing users: Participant observation, ethnography, Data collection	
	17	Analyzing, interpreting and presenting data	
	18	Qualitative analysis, Feeding the findings back into design	
7	19	Asking users and experts: Interviews	Mid Term Exam
	20	Asking users and experts: Questionnaires,	
	21	Asking users and experts: Inspections	
8	22	Asking users and experts: Inspections (Contd.)	
	23	Asking users and experts: walkthroughs.	
	24	Asking users and experts: walkthroughs (Contd.)	
9	25	Universal design: Universal design principles	
	26	Multi-modal interaction	
	27	Designing for diversity	
10	28	Task analysis: Differences between task analysis and other techniques, Task decomposition	Class Test 3
	29	Knowledge-based techniques, Entity-relationship-based techniques	
	30	Sources of information and data collection, Uses of task analysis.	
11	31	Modeling rich interaction: Status-event analysis, Rich contexts	
	32	Low intention and sensor	
	33	Low intention and sensor (Contd.)	
12	34	Ubiquitous computing and augmented realities: Ubiquitous computing applications research.	
	35	virtual and augmented reality	
	36	Information and data visualization	
13	37	Hypertext, multimedia and the world wide web: Understanding hypertext,	
	38	Understanding hypertext(Contd.)	
	39	Finding things	
14	40	Web technology and issues	
	41	Static web content	
	42	Dynamic web content	

ASSESSMENT STRATEGY

Components		Grading	CO	Blooms Taxonomy
Continuous Assessment (40%)	Test 1-2	20%	CO1	C1-C3
			CO2	C4, C6
	Class Participation	5%	CO4	A2
	Mid term	15%	CO2	C4, C6
Final Exam		60%	CO1	C1-C3
			CO3	C4
Total Marks		100%		

(CO = Course Outcome, C = Cognitive Domain, P = Psychomotor Domain, A = Affective Domain)

REFERENCE BOOKS

1. Julie A. Jacko (Ed.). (2012). Human-Computer Interaction Handbook (3rd Edition). CRC Press. ISBN 1-43

2. Alan Dix, Janet Finlay, Gregory Abowd, and Russell Beale (2003): Human-Computer Interaction. 3rd Ed Prentice Hall, 2003. <http://hcibook.com/e3/> ISBN 0-13- 046109-1

3. Yvonne Rogers, Helen Sharp, Jenny Preece (2019): Interaction Design: Beyond Human Computer Interaction (3rd Edition), John Wiley & Sons.

4. Schneiderman B. and Plaisant, C.: Designing the User Interface (5th Edition), Addison-Wesley. Jonathan Lazar, Jinjuan Heidi Feng, & Harry Hochheiser Research Methods in Human-Computer Interaction, Wiley, 2010. ISBN 0-470-72337-8, 978-0-470-72337-1

REFERENCE SITE

CSE-429: Computer Security

COURSE INFORMATION			
Course Code	: CSE-429	Lecture Contact Hours	: 3.00
Course Title	: Computer Security	Credit Hours	: 3.00
PRE-REQUISITE			
Course Code: Nil Course Title: Nil			
CURRICULUM STRUCTURE			
Outcome Based Education (OBE)			
RATIONALE			
The Computer Security course is designed to provide a comprehensive understanding to the modern security system in computer science. The course begins with the history and development of security. Then it deals with various security models, cryptography, security attacks and the fundamental security objectives. This course also motivates to gather brief review of computer crimes and causes, internet, strategies, crime prevention, etc.			
OBJECTIVE			
1. To understand the development of security, traditional encryption, security attacks and the fundamental security objectives. 2. To determine and analyse the security objectives, attacks, and models, so is able to recognize the security requirements in real-life cases.			